



Deep Generative models. ✓

: Graphical models + deep learning

Generate data

{ Text
Image
Video
Audio
Code

Notebook LM

kernel
probab
mach

Classroom Technology Support
Information Technology Services • UC San Diego
its.ucsd.edu

Energy is priceless.
Please turn projector off at end of class.

ng
Notebook LM

kernels.

Linear algebra

probability distributions

Optimization

machine learning

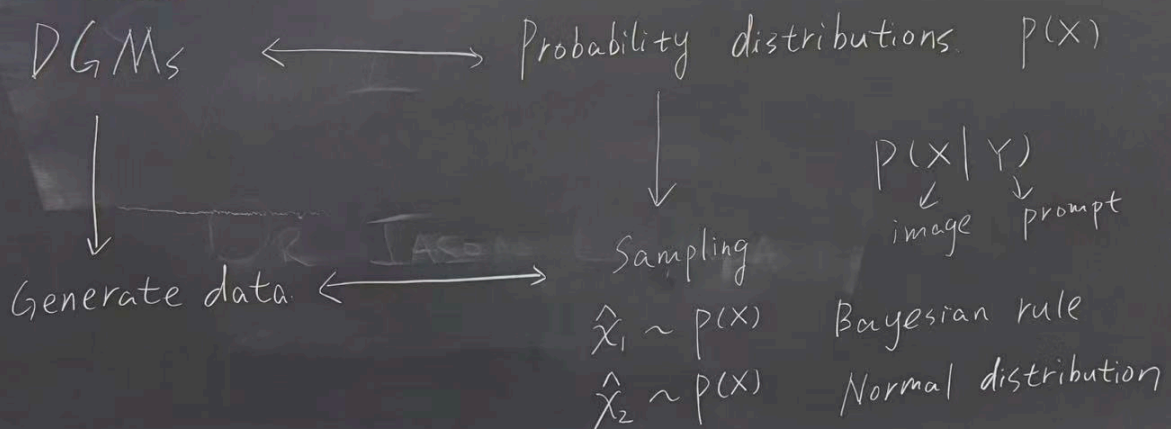
DGMs ←



Generate data

celess.

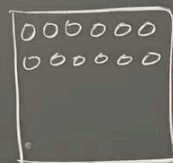
ff at end of class.



w_i
 $\downarrow$
 0
Rel
I
 w_i
nod
Bu
Do

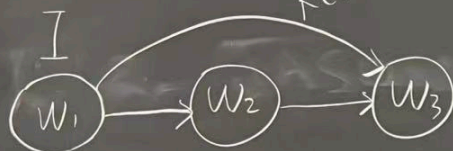
W_1 W_2 W_3 ...

↓ ↓ ↓
Objects



Relations

Relations



nodes

Graph

I ?

like 0.6

apple 0.1

at 0.05

Graph of random variables



probability graphical models

Build a graph.
Domain knowledge

Deep Generative models ✓

Graphical models + deep learning

$$P(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$

$$P(x; w) = f(x, w)$$

$$P(z|x)$$

$$P(\text{images} | \text{prompt})$$

$$P(x) \quad \underline{P(x|y)}$$

Key tasks in

① Design the

② Paramet

③ Data gen

④ Data un

Key tasks in DGMs

① Design the DGM

② Parameter learning

③ Data generation

④ Data understanding (inference)

Energy is priceless.

Please turn projector off at end of class.

Build the model

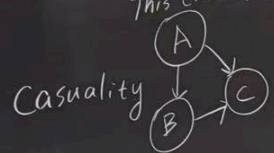
Use the model

Bayesian network

Graphical models

Directed
Undirected

This class is easy



I get an A

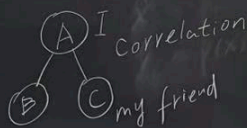
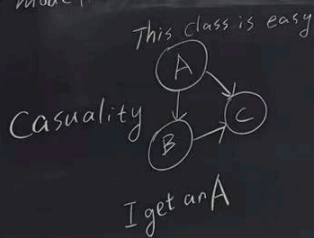
Energy is priceless.
Turn projector off at end of class.

Build the model

Bayesian network.

Graphical models { Directed graphical models (Bayesian network)
Undirected graphical models (Markov Random Fields)

Use the model.



W_1, W_2, W_3
↓ ↓ ↓
Objects

Relations

```
graph TD; W1((W1)) --> W2((W2)); W2((W2)) --> W3((W3));
```

I ?

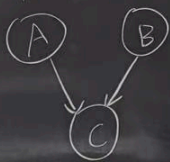
nodes Graph

Build a graph
Domain knowledge

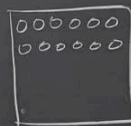
Graph of
probability gr

Bayesian network)

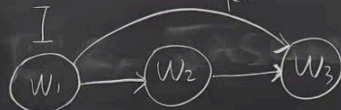
(Markov Random Fields)



W_1 W_2 W_3 ...
↓ ↓ ↓
Objects



Relations



nodes

Graph

Build a graph.
Domain knowledge

I ?

like 0.6
apple 0.1
at 0.05

Graph of random variables



probability graphical models.